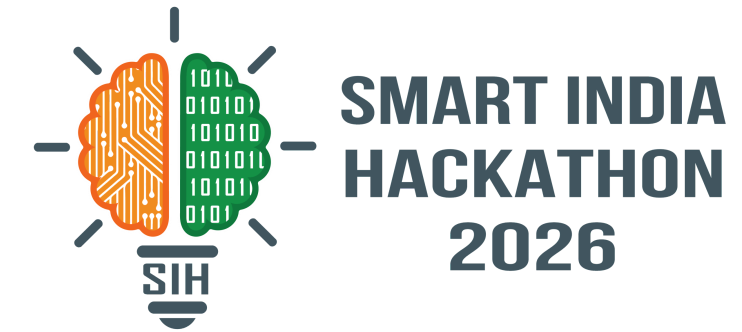
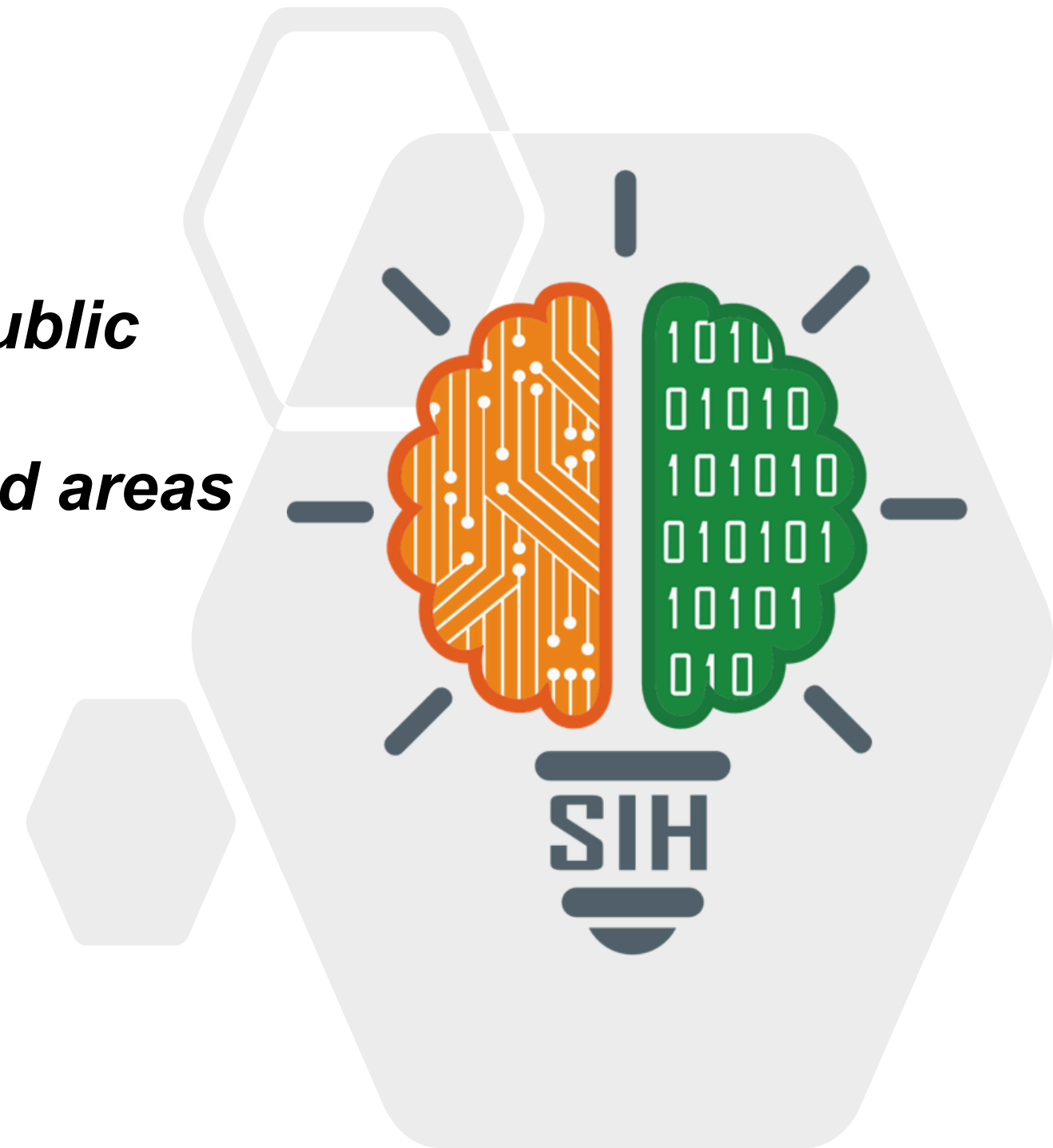


SMART INDIA HACKATHON 2026



- Problem Statement ID – ***SIH26133***
- Problem Statement Title - ***Accessibility and quality of public healthcare services, particularly in rural and underserved areas***
- Theme- ***MedTech / BioTech / HealthTech***
- PS Category- ***Software***
- Team ID-
- Team Name (Registered on portal) : ***LAKSHYA***



The Problem:

Geographic & Connectivity Isolation:



Zero-Bandwidth Environments: Rural Primary Health Centres (PHCs) frequently operate completely off-grid, isolated from standard internet access.



Blind Supply Chains: Critical medical supplies expire or stock-out unnoticed because neighbouring clinics cannot synchronize inventory data



Inaccessible Patient Data: Vital medical histories remain trapped in physical paper trails or unreachable cloud servers, leaving doctors blind during life-or-death triage.



Delayed Clinical Decision-Making: Frontline health workers often lack immediate access to clinical decision support, forcing them to rely on limited resources or wait for guidance from distant facilities—delaying treatment for high-risk patients.



High-Risk Patients Lost to Follow-Up: Patients requiring repeated monitoring—especially maternal, child and chronic-care patients—can become difficult to track after their initial visit, particularly when they move between facilities.

The Core Idea:

A 100% Offline-First Architecture We are building a decentralized network built on these technical pillars:



"Engraved" Offline QR Identities: Critical survival data is not hidden behind a web link; it is compressed and physically engraved directly into the QR code's pixels, allowing local cameras to decrypt life-saving history with zero bandwidth.



Local Nodes & Radio Sync: PHCs run a zero-latency local interface. Inventory updates are compressed into micro-strings and broadcasted via Long Range (LoRa) radio to a connected district hub, bypassing standard telecom grids



Automated Data Mules (Sneakernet): For severe regional network blackouts, offline clinics queue data locally. Visiting supply vans execute an automated Bluetooth handshake to extract encrypted logs, physically transporting the data to connected urban zones



Digital Triage (Offline Clinical Decision Support): Frontline workers can record symptoms, vitals and risk factors locally. An offline triage engine prioritizes patients by urgency and flags high-risk cases for immediate escalation.



Closed-Loop Referral & Follow-Up: Referrals are digitally tracked from PHC → higher centre → treatment → follow-up, ensuring patients don't disappear between facilities and enabling monitoring of high-risk cases



Assisted Teleconsultation (Connect Without Connectivity): When a live consultation is possible, local health workers act as the bridge - capturing clinical information offline and synchronizing consultation packets whenever connectivity becomes available

01.OFFLINE PHC – Capture & Store

- Engraved QR → Flutter/Dart → SQLite
- Patient's engraved QR is scanned locally to extract critical health data without Internet.
- Flutter/Dart provides the offline PHC interface for instant triage and medicine deduction.
- SQLite stores patient and inventory transactions locally, ensuring zero data loss during connectivity gaps.

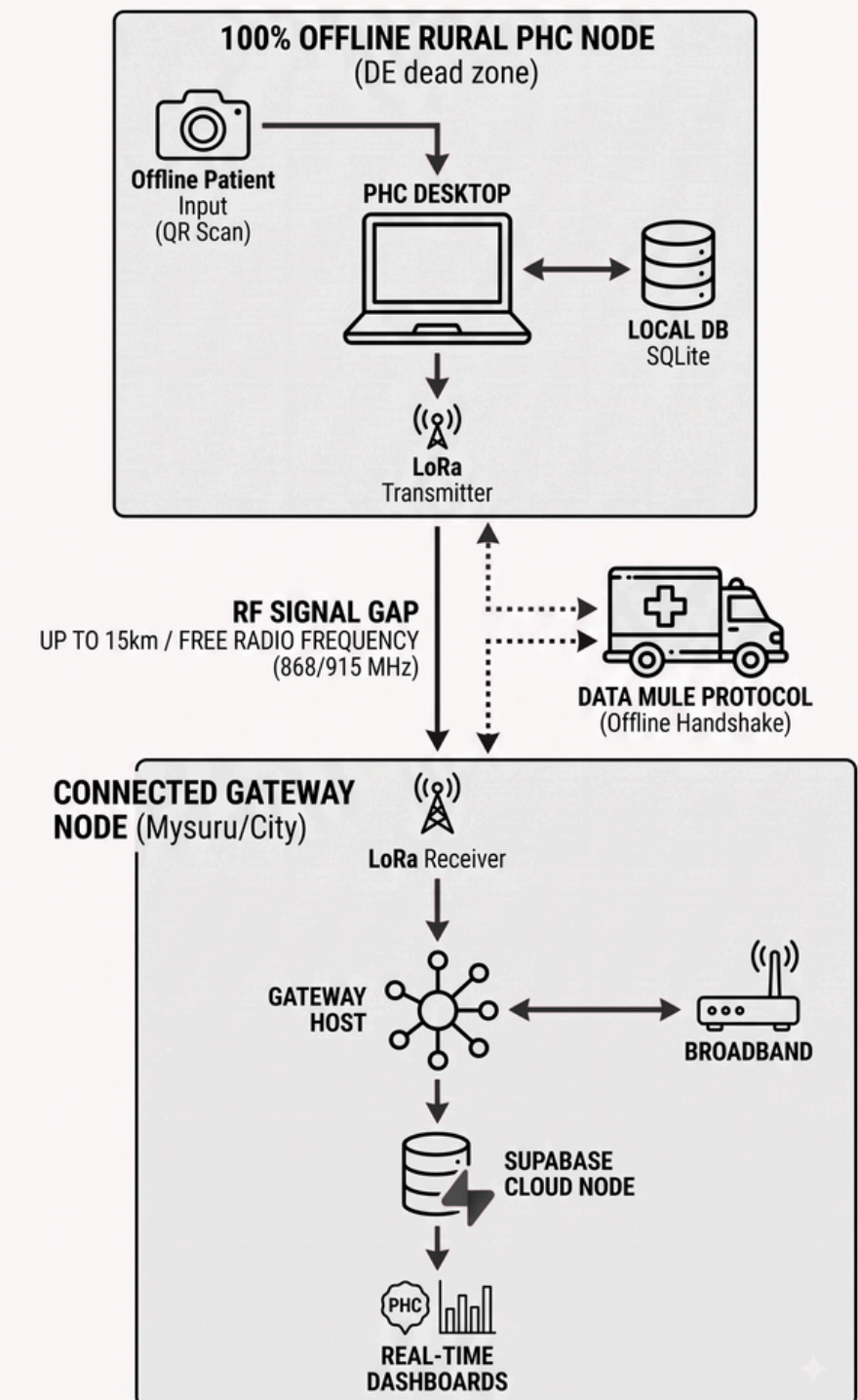
02.DEAD-ZONE BRIDGE – Compress & Transfer

- SQLite → Python → LoRa / Bluetooth
- Python middleware converts queued SQLite updates into encrypted 256-byte micro-strings.
- LoRa (868/915 MHz) transmits essential inventory data across ~15 km, bypassing cellular dead zones.
- If LoRa is unavailable, Bluetooth Data Mule transfers the cached data through existing ASHA workers/supply vehicles.

03.CONNECTED GATEWAY – Sync & Act

- LoRa Receiver → Supabase/PostgreSQL → Dashboard
- The City Gateway receives and decrypts incoming micro-strings.
- Supabase/PostgreSQL synchronizes PHC data to the cloud once connectivity is restored.
- District dashboards provide inventory visibility, enable surplus–shortage matching, and trigger restocking/emergency alerts.

SYSTEM ARCHITECTURE: BRIDGING THE ISOLATION



✓ Analysis of feasibility

100% Offline Operation:

- 1 SQLite enables PHCs to store and access critical data without Internet.

Low-Cost Infrastructure:

- 2 LoRa nodes provide long-range communication with minimal deployment cost.

Offline Patient Access:

- 3 Engraved QR enables retrieval of critical patient identity, history and allergy information

Existing Logistics Utilization:

- 4 ASHA workers and supply vehicles can act as Bluetooth data carriers.

! Potential Challenges

Hardware Failure:

- 1 LoRa nodes or local devices may malfunction in remote PHCs.

QR Damage/Wear:

- 2 Physical QR codes may become difficult to scan or decode.

Data Security Risk:

- 3 Sensitive patient information requires strong protection during storage and transfer.

Data Synchronization Conflicts:

- 4 Multiple PHCs may update inventory while remaining disconnected.

IMPACT



Zero-Connectivity Care

Critical medical information remains accessible even without a network



Faster Emergency Decisions

Offline roads, instant triage and rapid data transfer help healthcare workers make informed decisions



Smarter Medicine Distribution

Critical medical information remains accessible even without a network



BENEFITS



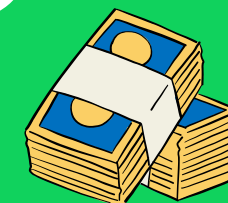
Social Benefit

Ensures access to critical patient information and essential healthcare services, even in remote, zero-connectivity regions



Government Benefit

Provides district authorities with actionable visibility into medicine shortages, patient needs and healthcare resources.



Economic Benefit

Reduces medicine wastage, prevents stock-outs and minimizes manual work through offline data synchronization

1. Official Portals & Technical Frameworks:

- Smart India Hackathon Portal: SIH 2026 Official Website — Source for problem statements, guidelines, and submission portals.

2. Ayushman Bharat Digital Mission (ABDM):

- ABDM Developer Sandbox — National digital health infrastructure framework used for interoperable health records and sandbox testing.

3. HL7 FHIR & SMART Health Cards:

- SMART Health Cards Specification — Interoperability standard utilized for encoding self-contained, cryptographically signed medical histories into offline QR codes.

4. National Telemedicine Backbone:

- eSanjeevani Government Portal — National telehealth benchmark analyzed during competitor evaluations.

5. Global Health Platforms:

- DHIS2 Software Platform — International open-source health information system reviewed for macro-level health data handling.

1. SQLite For Beginners - <https://www.youtube.com/watch?v=8Xyn8R9eKB8&pp=ygUfc3FsaXRlIGV4cGxhaW4gaW4gc2ltb2xIIHdvcmRzIA%3D%3D>

2. LoRa Technology - <https://youtu.be/ql4a9JHO2sc>

3. ASHA Workers - <https://www.youtube.com/watch?v=i0OG9P4fiRk&pp=ygUUQVNIQSB3b3JrZXJzICB2aWRlbyA%3D%3D>

4. PHC in India - <https://www.youtube.com/watch?v=mvljECmG00g&t=7s&pp=ygU6cHJpbWVyeSB0ZWVsdGggY2FyZSBpbjBpbmRpYSBjb21tdW5pdHkgaGVhbHRoIG51cnNpbmcgbmV3cw%3D%3D>

5. Ayushman Bharath Mission - <https://abdm.gov.in/>

6. PHC's Website - <https://aam.mohfw.gov.in/>

7. Indian Public Health Standards - <https://iphs.mohfw.gov.in/>

8. Ministry of Healthcare and Welfare - <https://mohfw.gov.in/>